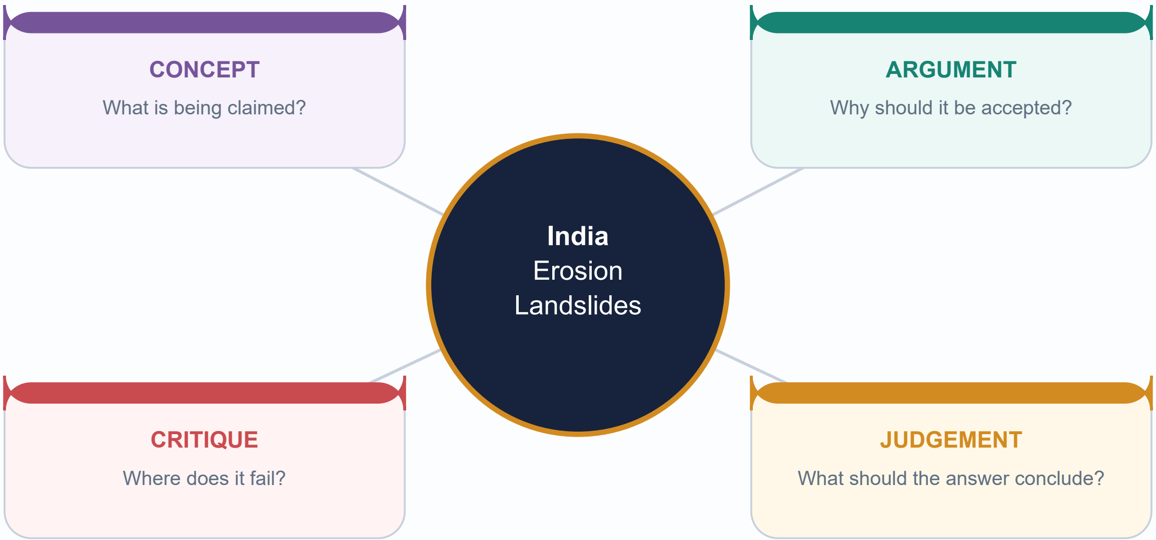
UPSC SOLVED PRACTICE WORKBOOK

India Erosion, Landslides & Groundwater — Solved Workbook

Geography 04 | Solved PYQs, hard MCQs, remedial loop and model Mains answers
Export date: 2026-08-15 | Original MCQs use strict A → B → C → D rotation
Mains answers: claim → named evidence → analysis → qualification → verdict



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

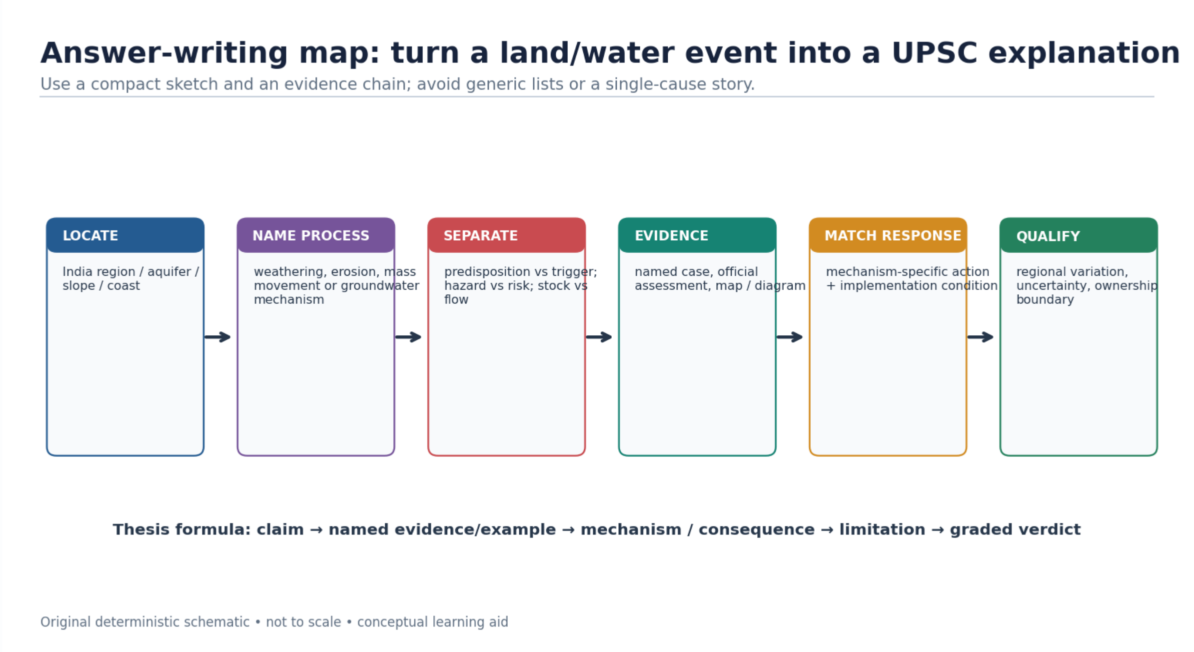
Workbook orientation: sources, key-status rule and answer standard

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

[FACT] Question wording/key status is controlled by local official-paper PDFs and routing ledgers. [LIMIT] Direct ownership is retained: 2018 urban harvesting is Topic 04; 2024 Gangetic food security is Geography Topic 36; 2021 landslides is Disaster Management; 2025 coastal intrusion is Environment; 2025 groundwater depletion is Economy.

WORKBOOK ANSWER STANDARD



The solution pattern applies to every Mains answer below.

Original deterministic schematic; conceptual, not to scale and not factual evidence.

STATIC THEORY

- [FACT] Local source priority: Geography Basic/Advanced 04 and related files; local book/PYQ PDFs; then dated official current sources. Qdrant was not required.

- [FACT] Mains solutions use **claim** → **named evidence/example** → **analysis** → **qualification** → **verdict** and end with **Why this earns marks**.

- [FACT] Official Prelims Set-A keys are locally available for 2024/2025 items used here. For 2018–2023, the routing ledger records no local official key; inferred answers remain explicitly labelled.

- [LIMIT] This workbook supplies physical Geography support to adjacent-owner PYQs but does not silently transfer Disaster Management, Environment or Economy ownership.

- [ANALYSIS] Original MCQs alone use strict A→B→C→D rotation; fixed official-PYQ option order is preserved and separately labelled.

MUST-KNOW FACTS

1. Question–answer separation is deliberate.
2. Named evidence must support an inference, not decorate it.

UPSC TRAPS

WRONG: Treating a local answer key absence as permission to omit a PYQ.

CORRECT: Retain it with the required inferred-answer label, confidence and elimination.

MAINS ANGLE

Before writing: decode directive, marks and word limit; then select evidence units and a visual only if it advances the demand.

STUDY LINK

Main learning session Sections 16–21.

HIGH

PYQ 1 — UPSC Mains 2018 GS-I Q14 (direct owner): urban water harvesting

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question (250 words): 'The ideal solution of depleting ground water resources in India is water harvesting system.' How can it be made effective in urban areas?

STATIC THEORY

- Demand: assess **effectiveness in urban areas**, not merely define water harvesting. Direct owner: Geography Advanced 04.

- Answer plan: urban depletion/context → capture + water quality → aquifer/drainage fit → demand management → governance/monitoring → qualified verdict.

MUST-KNOW FACTS

1. Verified from local official 2018 GS-I paper OCR.
2. No official answer key is relevant to a Mains answer.

UPSC TRAPS

WRONG: Writing a rural check-dam list.

CORRECT: Address impervious cover, roofs, drainage, contamination and urban demand.

MAINS ANGLE

Use a city-water-system diagram: roof/road runoff → quality control → storage/recharge → demand/reuse → monitoring.

STUDY LINK

Main Section 12; full model solution next.

HIGH

PYQ 1 — Model answer: urban harvesting effectiveness

GS-I + Prelims / bounded GS-III links Geography

STATIC THEORY

- [CLAIM] Urban water harvesting can reduce pressure on depleting groundwater only when it is designed as an aquifer-and-demand programme rather than as a construction target. [EVIDENCE] Impervious roofs and roads rapidly route monsoon runoff away from recharge, while dense urban abstraction can lower local water levels. [ANALYSIS] Capture at building and neighbourhood scale can either store water for non-potable use or recharge a suitable aquifer, reducing peak runoff and dependency where the water budget permits.

- [EVIDENCE] The Ministry of Jal Shakti’s 3 August 2026 groundwater-management release reports NAQUIM mapping/management-plan dissemination and conservation/recharge initiative status. [ANALYSIS] This makes hydrogeological screening possible: recharge wells must match transmissive strata, protect recharge zones and avoid pushing contaminated runoff into groundwater.

- [EVIDENCE] The 2025 national assessment vocabulary separates recharge, extractable resource and extraction. [ANALYSIS] Thus effectiveness must be measured through water-level/quality trends and reduced demand, not a count of pits or tanks.

- [QUALIFICATION] First-flush diversion, filtration, maintenance, storm-water-drain integration, wetland/tank protection, reuse, leakage control and pumping regulation are necessary; a recharge structure cannot compensate for unrestricted extraction or poor-quality inflow. [VERDICT] Cities should combine ward/aquifer mapping, enforceable building systems, public-space blue–green infrastructure and transparent monitoring. **Why this earns marks:** it answers effectiveness, uses a named official evidence base, connects urban hydrology to aquifer fit and includes a credible implementation limitation.

MUST-KNOW FACTS

1. Named evidence: Ministry of Jal Shakti 3 Aug 2026 / NAQUIM status; 2025 assessment vocabulary.

UPSC TRAPS

WRONG: Claiming a programme or structure automatically revived water levels.

CORRECT: Separate reported implementation/status from measured outcome.

MAINS ANGLE

250-word structure: thesis (2 lines) → 4 mechanism headings → condition/limitation → one-line verdict.

STUDY LINK

Direct 2018 GS-I route.

HIGH

PYQ 2 — UPSC Mains 2024 GS-I Q14 (Geography Topic 36 owner): Gangetic food security

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question (250 words): The groundwater potential of the Gangetic valley is on a serious decline. How may it affect the food security of India?

STATIC THEORY

- Demand: show a hydrology-to-food-security transmission mechanism. Registered owner: Geography Basic 36; Topic 04 supplies aquifer mechanism.

- Do not substitute a generic water-conservation list for food-security analysis.

MUST-KNOW FACTS

1. Verified from local official 2024 GS-I paper PDF, page 3.

UPSC TRAPS

WRONG: Equating food security with only grain tonnage.

CORRECT: Use availability, access, utilisation/quality and stability, with smallholder distributional effects.

MAINS ANGLE

Use the groundwater–irrigation–food-security chain; state the owner boundary in a study note, not in an exam answer.

STUDY LINK

Main Section 11; model solution next.

HIGH

PYQ 2 — Model answer: Gangetic groundwater and food security

GS-I + Prelims / bounded GS-III links Geography

STATIC THEORY

- [CLAIM] A serious decline in Gangetic-valley groundwater potential can weaken India's food security because a productive alluvial agricultural belt relies on groundwater as a seasonal irrigation buffer. [EVIDENCE] The 2024 GS-I question itself foregrounds the Gangetic valley; the Ministry's 2025 national assessment distinguishes recharge, extractable resource and extraction. [ANALYSIS] Where withdrawal persistently exceeds replenishment, water levels/well yields can fall and lifting water requires more energy and capital.

- [EVIDENCE] Tube-well irrigation supports dry-season reliability in the rice–wheat landscape. [ANALYSIS] Reduced reliability can delay irrigation, raise cultivation cost and expose yields to rainfall variability, affecting food **availability** and price/income **access**.

- [EVIDENCE] Smallholders typically have less capacity to deepen wells, pay higher energy bills or shift technology/crops. [ANALYSIS] The first impact can therefore be distributional: unequal access to irrigation translates into unequal production and purchasing power, while national supply stability faces regional stress.

- [EVIDENCE] Conjunctive use, crop diversification, efficient irrigation, recharge-zone/wetland protection and aquifer-level water budgeting address different links. [ANALYSIS] Demand-side crop, procurement and energy incentives matter alongside recharge.

- [QUALIFICATION] Groundwater irrigation has also raised productivity and drought resilience; the inference is not 'stop pumping', but govern abstraction/recharge equitably at aquifer scale. [VERDICT] Food security is protected when hydrological balance and farm incentives are jointly managed. **Why this earns marks:** it directly links aquifer stress to all food-security pillars, uses named evidence and smallholder analysis, and gives a balanced demand-side conclusion.

MUST-KNOW FACTS

1. Evidence density: Gangetic alluvium, 2024 PYQ, 2025 assessment vocabulary, tube wells, smallholders, conjunctive use.

UPSC TRAPS

WRONG: Saying only 'less groundwater means less crop'.

CORRECT: Explain cost, timing, equity and national stability transmission routes.

MAINS ANGLE

Sketch the five-box food-security chain; keep data dated and no unsupported Gangetic decline rate.

STUDY LINK

2024 GS-I direct demand / Topic 36 owner.

HIGH

PYQ 3 — UPSC Mains 2021 GS-I Q4 (adjacent Disaster Management owner): landslides

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question (150 words): Differentiate the causes of landslides in the Himalayan region and Western Ghats.

STATIC THEORY

- Demand: **differentiate**, so use a paired regional comparison plus shared anthropogenic/drainage qualifiers.

- Owner: Disaster Management Basic 10. Geography contribution: relief, materials, water, tectonic setting and spatial comparison.

MUST-KNOW FACTS

1. Verified from local official 2021 GS-I paper PDF, page 2.

UPSC TRAPS

WRONG: Writing only a common cause list.

CORRECT: Contrast setting-specific predispositions while retaining shared triggers/amplifiers.

MAINS ANGLE

150 words: Himalaya (3–4 distinctions) | Western Ghats (3–4 distinctions) | shared human/drainage qualification.

STUDY LINK

Main Sections 5–6; model solution next.

HIGH

PYQ 3 — Model answer: Himalaya versus Western Ghats landslide causes

GS-I + Prelims / bounded GS-III links Geography

STATIC THEORY

- [CLAIM] Both regions experience gravity-driven slope failure, but their predispositions differ because their tectonic history, relief, material and rainfall–drainage settings differ. [EVIDENCE] The Himalaya is a young collision belt with steep relief, active deformation/seismicity, fractured rocks, deep river incision and widespread weathered/debris slopes. [ANALYSIS] Intense/prolonged monsoon rain or seismic shaking can therefore act on already stressed material; road cutting can remove toe/lateral support and concentrate drainage.

- [EVIDENCE] The Western Ghats and Nilgiris are older escarpment/plateau-margin landscapes where deeply weathered or lateritic/altered material, short intense orographic rain, slope cutting, quarrying, drainage change and land-use pressure can be important. [ANALYSIS] This does not make them tectonically identical to the Himalaya; material strength and water pathways require local diagnosis.

- [EVIDENCE] GSI susceptibility mapping and ISRO’s Landslide Atlas provide official spatial inventory/mapping context. [ANALYSIS] They reinforce that hazard assessment combines terrain, material, rain and exposure rather than one-factor labels.

- [QUALIFICATION] Rainfall is commonly an immediate trigger, but neither rainfall nor road construction alone establishes causation for an individual failure. [VERDICT] Region-specific zonation, drainage and slope design, regulated cutting/land use and warning/evacuation should be adapted to site evidence. **Why this earns marks:** it obeys ‘differentiate’, gives named regional evidence and official mapping sources, explains shared drivers and avoids deterministic attribution.

MUST-KNOW FACTS
1. Himalaya: young collision/high relief/fractures/seismicity. Western Ghats: weathered escarpment/orographic rain/drainage/land use.

UPSC TRAPS
WRONG: Calling the Western Ghats a young active collision belt.

CORRECT: Differentiate geology while recognising local structural and material variability.

MAINS ANGLE
Mini-table with headings: setting, material/relief, water/trigger, human amplification, response.

STUDY LINK
Adjacent DM owner retained.

HIGH

PYQ 4 — UPSC Mains 2025 GS-III Q8 (adjacent Environment owner): coastal intrusion

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question (150 words): Seawater intrusion in the coastal aquifers is a major concern in India. What are the causes of seawater intrusion and the remedial measures to combat this hazard?

STATIC THEORY

- Demand: causes **and** remedies; lead with freshwater-head / saline-interface mechanism.

- Owner: Environment and Ecology Basic 24. Geography contribution: aquifer/interface mechanism and coastal spatial reasoning.

MUST-KNOW FACTS
1. Verified from local official 2025 GS-III paper PDF, page 3.

UPSC TRAPS
WRONG: Treating seawater intrusion as surface flooding.

CORRECT: Explain landward/upward subsurface saline-interface movement.

MAINS ANGLE
150 words: mechanism (2 lines) → causes (3 grouped) → remedies (demand/recharge/barrier/monitoring) → site-specific qualification.

STUDY LINK
Main Section 13; model solution next.

HIGH

PYQ 4 — Model answer: coastal intrusion causes and remedies

GS-I + Prelims / bounded GS-III links Geography

STATIC THEORY

- [CLAIM] Seawater intrusion occurs when freshwater hydraulic head in a coastal aquifer becomes too low to resist saline-water movement; excessive pumping is commonly the central controllable driver. [EVIDENCE] A coastal well can create a local drawdown cone; saline water may move landward along the interface or rise beneath the well as upconing. [ANALYSIS] The process is subsurface and differs from visible storm-surge flooding.

- [EVIDENCE] Over-abstraction for irrigation, industry or urban supply lowers freshwater head; paved/altered recharge areas reduce replenishment; coastal geology, sea-level/storm stresses and poorly managed saline pathways can compound vulnerability. [ANALYSIS] Their relative importance varies by aquifer, so a universal causal ranking is unsafe.

- [EVIDENCE] The groundwater-management approach—monitoring, aquifer mapping and recharge planning—provides a decision framework. [ANALYSIS] Remedies should reduce/redistribute pumping, protect recharge zones, use safe managed recharge where hydrogeologically suitable, consider site-specific hydraulic/subsurface barriers, monitor salinity and head, and adapt water/crop/land use.

- [QUALIFICATION] Desalination may improve supplied water but does not itself restore freshwater head; unrestricted injection may worsen quality if source water/aquifer fit is ignored. [VERDICT] Coastal aquifer security needs monitored demand management plus recharge/protection at the aquifer scale. **Why this earns marks:** it gives the mechanism, groups causes and remedies, distinguishes flooding from intrusion and ends with an implementable qualification.

MUST-KNOW FACTS

1. Named evidence: interface/upconing mechanism; groundwater-management planning context.

UPSC TRAPS

WRONG: Offering only desalination.

CORRECT: Treat it as a supply option, not a substitute for aquifer-head management.

MAINS ANGLE

Draw a freshwater wedge and pumping well; label schematic/not to scale.

STUDY LINK

Adjacent Environment owner retained.

HIGH

PYQ 5 — UPSC Mains 2025 GS-III Q13 (adjacent Economy owner): depletion and government steps

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question (250 words): Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater?

STATIC THEORY

- Demand: factors **and** state steps. Owner: Economy Basic 14. This solution supplies physical-geography evidence without replacing the agriculture/economic-policy owner.

- Use dated assessment/programme status but do not turn announced implementation into outcome claims.

MUST-KNOW FACTS
1. Verified from local official 2025 GS-III paper PDF, page 3.

UPSC TRAPS
WRONG: Writing only natural drought causes.

CORRECT: Include cropping, energy, urban/industrial demand, aquifer/recharge conditions, land-use and governance.

MAINS ANGLE
250 words: diagnosis by demand/recharge/governance → government instruments → outcomes/limits → verdict.

STUDY LINK
Main Sections 10, 12, 15; Economy owner retained.

HIGH

PYQ 5 — Model answer: depletion factors and government steps

GS-I + Prelims / bounded GS-III links Geography

STATIC THEORY

- [CLAIM] Groundwater depletion results when withdrawal persistently exceeds recharge at the relevant aquifer scale; India's drivers combine hydrogeology with agricultural, urban and energy-policy choices. [EVIDENCE] The Ministry of Jal Shakti 5 February 2026 release reports separate 2025 national estimates of recharge (448.52 BCM), extractable resource (407.75 BCM) and extraction (247.22 BCM). [ANALYSIS] Their distinction shows why national pumping cannot be read as a local sustainability verdict.

- [EVIDENCE] Water-intensive cropping, assured irrigation demand, subsidised/low-cost energy, tube-well access, urban/industrial demand, leakage and sealed recharge areas can raise abstraction or reduce infiltration. [ANALYSIS] In hard-rock settings limited storage/fracture pathways and in alluvial/coastal systems depth/quality heterogeneity sharpen local vulnerability.

- [EVIDENCE] CGWB monitoring, NAQUIM aquifer maps/management-plan dissemination, groundwater assessment, rainwater/artificial-recharge guidance, Jal Shakti Abhiyan/participatory approaches and Atal Bhujal pilot learning are reported policy instruments. [ANALYSIS] They support diagnosis, recharge-zone identification, community water budgets and demand planning rather than a one-size-fits-all structure.

- [QUALIFICATION] Government steps work only with state/local regulation, crop/energy incentive reform, water-quality safeguards, maintenance and transparent level/quality outcomes; structure counts are not recovery proof. [VERDICT] India should couple aquifer mapping and recharge with demand-side irrigation/urban reforms and equitable local governance. **Why this earns marks:** it answers both parts, uses dated official evidence, links physical and incentive drivers, preserves outcome caution and offers a matched policy mix.

MUST-KNOW FACTS
1. Evidence density: assessment, CGWB/NAQUIM, recharge programmes, Atal Bhujal, crop/energy/urban drivers.

UPSC TRAPS
WRONG: Claiming all government schemes restored every aquifer.

CORRECT: Use status/implementation language unless measured outcome is verified.

MAINS ANGLE
Name the adjacent Economy owner in revision notes, but answer the substantive demand in the exam.

STUDY LINK
Adjacent Economy owner retained.

HIGH

PYQ 6 — UPSC Prelims 2024 GS-I Q7 (official Set-A key available)

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: Statement-I: Rainfall is one of the reasons for weathering of rocks. Statement-II: Rain water contains carbon dioxide in solution. Statement-III: Rain water contains atmospheric oxygen. Which option is correct? A Both II and III are correct and both explain I. B Both II and III are correct, but only one explains I. C Only one of II and III is correct and explains I. D Neither II nor III is correct.

STATIC THEORY

- **OFFICIAL SET-A KEY VERIFIED LOCALLY: A.** Rainwater can contain dissolved CO2, supporting carbonic-acid/carbonation pathways, and dissolved atmospheric O2, supporting oxidation pathways. Thus both statements are correct and each can contribute to weathering.

- Elimination: do not reduce weathering to rainfall alone—mechanical, mineralogical, structural and climatic controls remain—but the question asks whether rainwater composition can explain its role.

MUST-KNOW FACTS

1. Official 2024 Set-A key scan records Q7 = A.

2. Question text verified from local 2024 GS-I Set-A paper, page 5.

UPSC TRAPS

WRONG: Assuming only CO2 matters because chemical weathering is mentioned.

CORRECT: Test oxygen/oxidation independently; both statements support Statement I.

MAINS ANGLE

Prelims bridge: distinguish a true contributing mechanism from a claim that it is the sole mechanism.

STUDY LINK

Main Sections 2–3.

HIGH

PYQ 7 — UPSC Prelims 2025 GS-I Q28 (official Set-A key available)

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: Statement I: In the context of effect of water on rocks, chalk is a very permeable rock whereas clay is quite impermeable or least permeable. Statement II: Chalk is porous and hence can absorb water. Statement III: Clay is not at all porous. Which option is correct? A Both II and III are correct and both explain I. B Both II and III are correct but only one explains I. C Only one of II and III is correct and explains I. D Neither II nor III is correct.

STATIC THEORY

- **OFFICIAL SET-A KEY VERIFIED LOCALLY: C.** Statement II is correct in the question's intended contrast: chalk's pore/fracture network permits absorption/transmission. Statement III is false because clay can have pore space even though its very small/poorly connected pores make it poorly permeable.

- Elimination: porosity is storage space; permeability concerns connected flow. Never treat 'impermeable' as 'no pores at all'.

MUST-KNOW FACTS

1. Official 2025 Set-A key scan records Q28 = C.

2. Question text verified from local 2025 GS-I Set-A paper, page 13.

UPSC TRAPS

WRONG: Equating low permeability with zero porosity.

CORRECT: Clay demonstrates why the two properties must be separated.

MAINS ANGLE

Use the same distinction for aquifer versus aquitard answers.

STUDY LINK

Main Sections 8–9.

HIGH — PYQ 8 — UPSC Prelims 2021 GS-I Q64 (official key unavailable locally)

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: The black cotton soil of India has been formed due to the weathering of: A brown forest soil; B fissure volcanic rock; C granite and schist; D shale and limestone.

STATIC THEORY

- **INFERRED ANSWER — NOT OFFICIALLY VERIFIED. Confidence: High. Answer B (fissure volcanic rock).** Black/regur soil is associated with weathering of Deccan basalt, a fissure/flood-basalt volcanic province.

- Elimination: granite/schist may yield other soil materials; shale/limestone is not the standard parent-material cue; 'brown forest soil' is a soil category, not the named rock source. The repository holds no official 2021 Prelims Set-A key, so B is not presented as official.

MUST-KNOW FACTS

1. Question text verified from local 2021 official Prelims paper, page 29.

2. Central routing ledger records 2018–2023 official keys unavailable locally.

UPSC TRAPS

WRONG: Calling black soil a product of alluvium or black minerals.

CORRECT: Use Deccan basalt parent material plus clay/shrink–swell properties.

MAINS ANGLE

Prelims bridge: parent material → weathering/product → property → crop/management consequence.

STUDY LINK

Main Section 2; local source/key status rule.

HIGH — PYQ 9 — UPSC Prelims 2023 GS-I Q76 (official key unavailable locally)

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: Statement-I: According to the United Nations World Water Development Report, 2022, India extracts more than a quarter of the world's groundwater withdrawal each year. Statement-II: India needs to extract more than a quarter of the world's groundwater each year to satisfy drinking water and sanitation needs of almost 18% of the world's population living in its territory. Which option is correct? A both correct and II explains I; B both correct but II does not explain I; C I correct but II incorrect; D I incorrect but II correct.

STATIC THEORY

- **INFERRED ANSWER — NOT OFFICIALLY VERIFIED. Confidence: High. Answer C.** Statement I is the report-based claim tested by the question. Statement II is not a correct explanation because India's groundwater withdrawal is driven substantially by uses beyond drinking water and sanitation, notably irrigation; the stated need cannot explain the total extraction share.

- Elimination: test the explanation clause, not only the population percentage. The local repository has the official 2023 question paper but no official key; this answer is an informed inference, not an official result.

MUST-KNOW FACTS

1. Question text verified from local 2023 official Prelims paper, page 31.

2. Key status: unavailable locally under central routing ledger.

UPSC TRAPS

WRONG: Assuming a true population share explains a groundwater-withdrawal share.

CORRECT: Check sectoral use and causal relevance.

MAINS ANGLE

Prelims bridge: water-use sector logic prevents a demographic-statistic distractor from becoming a causal explanation.

STUDY LINK

Main Sections 10–11; local source/key status rule.

HIGH

Original hard MCQs 1–8 — strict authored rotation A → B → C → D

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

[ANALYSIS] These are original questions. Correct keys rotate exactly A, B, C, D, A, B, C, D.

STATIC THEORY

- 1. **Key A.** Which definition is correct? A Weathering is in-situ breakdown/alteration; erosion includes removal/transport. B Erosion is in-situ only. C Regolith is necessarily soil. D Mass movement needs a river. **Explanation:** A preserves all process boundaries.

- 2. **Key B.** Which best explains why clay can be a poor aquifer material? A It has no pore space. B Pores can be small/poorly connected, so permeability is low. C It cannot contain water. D It is always fractured. **Explanation:** B distinguishes porosity from permeability.

- 3. **Key C.** Which action directly reduces gully headward advance? A Deepening bare channels. B Moving sediment downstream only. C Upslope runoff control plus gully-head stabilisation. D Pumping groundwater. **Explanation:** C acts on concentrated-flow energy and head erosion.

- 4. **Key D.** A slope is most likely to lose stability when: A vegetation label changes alone. B all rainfall stops. C a hazard map is issued. D saturation raises pore pressure/weight while toe support is removed. **Explanation:** D captures mechanics.

- 5. **Key A.** The best regional statement is: A Himalayan landslide risk combines young high relief/material, rainfall, seismicity, drainage and human modification. B every failure is seismic. C Western Ghats have no landslides. D all India slopes are alike. **Explanation:** A is multi-factor and regional.

- 6. **Key B.** Which is the safest Joshimath formulation? A One cause has been finally proven. B Integrated evidence is needed across ground material, drainage, slope and loading. C Subsidence always means a slide. D a satellite trend proves the full mechanism. **Explanation:** B preserves evidence limits.

- 7. **Key C.** Which statement about groundwater is correct? A Recharge equals extraction. B Water table is fixed. C Aquifer management needs recharge, withdrawal and quality evidence. D a map proves recovery. **Explanation:** C is stock-flow/quality logic.

- 8. **Key D.** Which coastal remedy is most complete? A desalination alone. B only a sea wall. C unrestricted recharge injection. D monitor head/salinity, curb pumping, protect/suitably recharge aquifer and use site-specific barriers where viable. **Explanation:** D matches mechanism.

MUST-KNOW FACTS

1. Keys 1–8: A B C D A B C D.

2. Each explanation identifies the precise category error avoided.

UPSC TRAPS

WRONG: Choosing the option with more generic environmental words.

CORRECT: Choose the option that contains the relevant physical mechanism and condition.

MAINS ANGLE

Use every wrong-option pattern as a one-line trap in revision.

STUDY LINK

Main Sections 2–15.

HIGH

Original hard MCQs 9–16 — strict authored rotation A → B → C → D

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

[ANALYSIS] Continuation of the original set. Keys rotate exactly A, B, C, D, A, B, C, D.

STATIC THEORY

- 9. **Key A.** Gangetic groundwater decline can affect food security first through: A irrigation reliability, pumping cost and farm output/income risk. B a change in coastal tide only. C disappearance of all alluvium. D an automatic arsenic diagnosis. **Explanation:** A gives the direct transmission route.

- 10. **Key B.** Which statement is correct for a confined aquifer? A Its upper boundary is necessarily the water table. B Water may rise in a well to a potentiometric level under pressure. C It cannot be depleted. D all wells must flow. **Explanation:** B describes confined head correctly.

- 11. **Key C.** Which quality statement is defensible? A Salinity always comes from sea. B Nitrate always has one source. C A water-quality claim needs sample/location/pathway evidence. D quantity and quality are identical. **Explanation:** C preserves source discipline.

- 12. **Key D.** A city recharge intervention is most defensible when it includes: A structure count only. B roof water without first flush. C unrestricted runoff injection. D hydrogeological fit, water-quality safeguards, demand reduction and monitoring. **Explanation:** D is a full effectiveness chain.

- 13. **Key A.** Which is a valid erosion-control principle? A Reduce runoff energy and protect surface cover before treating downstream symptoms. B Create longer bare slopes. C concentrate all flow. D remove vegetation on gullies. **Explanation:** A works at source.

- 14. **Key B.** The best distinction is: A hazard and risk are synonyms. B risk combines hazard with exposure/vulnerability. C trigger and predisposition are the same. D atlas equals prediction. **Explanation:** B is the standard risk framing.

- 15. **Key C.** Which is the best use of the 5 Feb 2026 assessment values? A apply them to a named well. B treat extraction as recharge. C date/attribute the national figures and separately analyse local aquifer drivers. D claim every district is safe. **Explanation:** C respects scale.

- 16. **Key D.** A strong 15-marker on coastal intrusion should: A list coast states. B discuss CRZ only. C call it surface flooding. D explain freshwater head/interface, group causes/remedies, and qualify site variation. **Explanation:** D matches directive and mechanism.

MUST-KNOW FACTS

1. Keys 9–16: A B C D A B C D.

2. Full authored 16-question sequence has no consecutive correct option.

UPSC TRAPS

WRONG: Using an official-PYQ key sequence to audit original practice.

CORRECT: Audit original questions only; official wording/options remain fixed.

MAINS ANGLE

Convert any MCQ explanation into a five-word Mains evidence cue.

STUDY LINK

Main Sections 8–16.

HIGH

Remedial MCQs 17–24 — strict authored rotation A → B → C → D

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

[ANALYSIS] These target frequent errors. Keys continue A, B, C, D, A, B, C, D.

STATIC THEORY

- 17. **Key A.** A well becomes artesian-flowing only when: A its confined-aquifer potentiometric level is above ground. B it is deep. C it is coastal. D it reaches clay. **Explanation:** A is the conditional pressure criterion.

- 18. **Key B.** A saline coastal well proves: A a cyclone flooded it. B only that salinity exists; intrusion mechanism needs head/geology/pumping evidence. C all city wells are saline. D no recharge is possible. **Explanation:** B avoids correlation-to-causation.

- 19. **Key C.** Which is a correct Joshimath evidence statement? A Any crack names the cause. B Construction is irrelevant. C Monitoring locates/records change but integrated studies are needed for attribution. D subsidence is always a landslide. **Explanation:** C is evidence discipline.

- 20. **Key D.** Which statement correctly handles rainfall and landslides? A Rain always causes slides. B Rain is irrelevant. C only antecedent rock matters. D rainfall may trigger failure on a predisposed slope; mechanism needs site evidence. **Explanation:** D separates trigger/predisposition.

- 21. **Key A.** Which term names water entering the ground surface? A infiltration. B percolation only. C discharge. D artesian flow. **Explanation:** A is entry; further downward movement is percolation.

- 22. **Key B.** A recharge structure's outcome should be evaluated with: A construction photograph alone. B level/quality trends, demand context, aquifer fit and maintenance evidence. C scheme name alone. D rainfall name alone. **Explanation:** B tests performance.

- 23. **Key C.** Which answer conclusion is best? A a one-cause certainty. B a scheme announcement equals success. C a mechanism-matched, monitored, region-sensitive intervention with qualification. D no intervention is possible. **Explanation:** C is a graded verdict.

- 24. **Key D.** Which statement is unsuitable in a UPSC answer without evidence? A water table can vary. B coastal aquifers may be vulnerable. C roads can alter drainage. D a precise national landslide rank/casualty/depth quoted from memory. **Explanation:** D violates factual discipline.

MUST-KNOW FACTS

1. Keys 17–24: A B C D A B C D.

2. All 24 authored workbook MCQs rotate exactly A → B → C → D.

UPSC TRAPS

WRONG: Memorising a key without reading the 'why'.

CORRECT: Classify the error: boundary, scale, mechanism, conditionality or evidence.

MAINS ANGLE

If a remedial answer is wrong, redraw the corresponding original schematic before retrying.

STUDY LINK

Main diagrams 01–13.

HIGH

Original Mains practice — 10 marks (150 words): integrated slope corridor

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: Distinguish weathering, erosion and mass movement. Show how their interaction can create risk along a Himalayan road corridor.

STATIC THEORY

- [CLAIM] Weathering is in-situ breakdown/alteration; erosion removes/transports material by an external agent; mass movement is gravity-driven downslope transfer. [EVIDENCE] Himalayan high relief, fractured rock and weathered regolith create susceptible material. [ANALYSIS] Road cuts can steepen/undercut a slope and change drainage; intense rain can saturate debris, increasing driving stress and pore pressure. A slide can block a road even without fluvial transport as the immediate movement. [EVIDENCE] GSI mapping and ISRO's Landslide Atlas show why terrain/material/rainfall information matters. [QUALIFICATION] Neither rain nor a road proves cause at a particular site; bedding, toe support and drainage require investigation. [VERDICT] Design needs site investigation, drainage/slope work and maintenance. **Why this earns marks:** it distinguishes all terms, applies them to a named India setting, uses official mapping evidence and qualifies causation.

MUST-KNOW FACTS

1. Evidence units: Himalayan corridor; GSI; ISRO.

UPSC TRAPS

WRONG: Turning a differentiate question into a definition-only answer.

CORRECT: Apply the distinctions to one linked corridor mechanism.

MAINS ANGLE

Use one arrow diagram only: weathering → regolith + cut/drainage → saturation → slide → road disruption.

STUDY LINK

Main Section 19.

HIGH

Original Mains practice — 15 marks (250 words): watershed-scale management

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: Examine why soil erosion and slope failure should be managed at watershed scale in India. Illustrate with distinct physiographic settings.

STATIC THEORY

- [CLAIM] Erosion and slope failure are watershed problems because cover, slope length, drainage and infiltration govern runoff energy, sediment transfer, pore-water conditions and downstream exposure. [EVIDENCE] The Chambal ravines demonstrate the sheet–rill–gully–ravine route. [ANALYSIS] Cover, contouring and gully-head control lower detachment and concentrated-flow energy before sediment is routed downslope.

- [EVIDENCE] Himalayan corridors combine high relief, fractured/weathered material, road cuts and intense rain. [ANALYSIS] Drainage, toe support, bioengineering, retaining measures and siting must work together; a downstream check structure cannot stabilise an unsafe cut slope.

- [EVIDENCE] Western Ghats/Nilgiris add deeply weathered material, orographic rain and land-use/drainage change. [ANALYSIS] They need local material/water-pathway diagnosis, not a copied Himalayan remedy.

- [EVIDENCE] CGWB aquifer-mapping/assessment language highlights infiltration/recharge links. [ANALYSIS] Catchment treatment can support soil moisture/recharge when it does not introduce quality/sediment problems.

- [QUALIFICATION] Watershed scale does not solve tenure, incentives, maintenance or equity by itself. [VERDICT] Use nested plot–slope–subcatchment–aquifer planning. **Why this earns marks:** it examines the proposition, supplies four named evidence units, differentiates settings and ends with a governance qualification.

MUST-KNOW FACTS

1. Use 4–6 evidence units in a 15-marker.

UPSC TRAPS

WRONG: A generic 'afforestation and awareness' conclusion.

CORRECT: Map each action to runoff, sediment, drainage, strength, recharge or exposure.

MAINS ANGLE

Comparison matrix + watershed arrow diagram.

STUDY LINK

Main Section 20.

HIGH

Original Mains practice — 20 marks (250 words): groundwater stock, food and coast

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

Question: 'Groundwater is a stock governed by flows, quality and institutions—not an invisible reserve waiting to be pumped.' Analyse this proposition for India, including food security and coastal aquifers.

STATIC THEORY

- [CLAIM] Groundwater is dynamic: recharge adds to storage, pumping/springs/baseflow remove it, and usable water depends on aquifer transmission and quality. [EVIDENCE] Ministry of Jal Shakti's 5 Feb 2026 release reports 2025 recharge 448.52 BCM, extractable resource 407.75 BCM and extraction 247.22 BCM. [ANALYSIS] The three figures are not interchangeable and cannot substitute for local aquifer diagnosis.

- [EVIDENCE] The 2024 Gangetic-valley PYQ connects decline to food security. [ANALYSIS] Recharge–withdrawal imbalance can increase lift/energy costs and reduce irrigation reliability, affecting food availability, access and stability; smallholders cannot adapt equally.

- [EVIDENCE] The 2025 coastal-intrusion PYQ shows that excessive pumping lowers freshwater head and shifts saline interface/upconing. [ANALYSIS] Salinity is a quality–quantity interaction, not surface flooding; desalination alone does not restore aquifer head.

- [EVIDENCE] NAQUIM, CGWB monitoring and participatory/management initiatives are institutional tools. [ANALYSIS] They must be combined with demand/crop/energy reform, recharge-zone protection, quality safeguards, conjunctive use and transparent monitoring.

- [QUALIFICATION] Maps, schemes and structure counts are not outcomes; local hydrogeology, maintenance and equity vary. [VERDICT] Manage groundwater as a monitored common-pool stock, not a supply-only asset. **Why this earns marks:** it uses dated official evidence, cross-links food/coast, distinguishes stock/flow/quality and finishes with a qualified institutional conclusion.

MUST-KNOW FACTS

1. Use 5–8 evidence units in a 20-marker; diagram should clarify stock-flow/interface links.

UPSC TRAPS

WRONG: Claiming recharge alone makes pumping sustainable.

CORRECT: Address demand, quality, scale and institutional conditions.

MAINS ANGLE

Two compact diagrams: aquifer stock-flow and coastal interface/food chain.

STUDY LINK

Main Section 21.

HIGH

Workbook final revision sheet — one-page recall and answer audit

GS-I + Prelims / bounded GS-III links Geography

INTRO & ORIGIN

[ANALYSIS] Final revision consolidates direct PYQ demands, category boundaries, India anchors and the evidence standard.

STATIC THEORY

- Definitions: weathering in situ; erosion transport; mass movement gravity; subsidence ground lowering; hazard physical propensity; risk hazard + exposure + vulnerability.

- Erosion: sheet → rill → gully → ravine; Chambal cue; cover/slope-length/drainage/gully-head response.

- Landslide: material + slope + water + trigger + exposure. Himalaya differs from Western Ghats; never one-cause attribution.

- Joshimath: old debris/ground conditions + drainage/water + slope + loading; monitoring ≠ settled causation.

- Groundwater: infiltration → percolation → recharge; aquifer vs aquitard; unconfined/confined/artesian; porosity vs permeability; hard-rock vs alluvial variability.

- 2025 assessment (5 Feb 2026): recharge 448.52 BCM; extractable 407.75 BCM; extraction 247.22 BCM—three distinct quantities.

- Gangetic food chain: balance → irrigation cost/reliability → output/income → availability/access/stability; qualify tube-well productivity benefits.

- Coastal intrusion: pumping/head decline → wedge/upconing; remedy demand + suitable recharge + barrier/monitoring; wedge ≠ flooding.

- PYQ boundary: direct 2018 urban harvest; 2024 Gangetic owner Topic 36; 2021 DM; 2025 Q8 Environment; 2025 Q13 Economy. 2024/2025 keys official locally; 2021/2023 inferred labels mandatory.

- Answer audit: directive met? 2–3/4–6/5–8 named evidence units for 10/15/20? mechanism explained? limit stated? final 'Why this earns marks' present?

MUST-KNOW FACTS

1. Original MCQ rotation audit: Q1–24 keys = A B C D repeated six times.

2. All Mains model answers end with **Why this earns marks**.

UPSC TRAPS

WRONG: Treating a source/status announcement as an outcome.

CORRECT: State implementation status and require measured level/quality/maintenance evidence.

WRONG: Ending with an unqualified generic solution.

CORRECT: End with a mechanism-matched, regional and evidence-limited verdict.

MAINS ANGLE

Last 30 seconds: claim → evidence → mechanism → qualification → verdict.

STUDY LINK

Revisit original assets 01–13 and direct PYQ solutions.

End of Register Notes - UPSC Agent / Copilot CLI